Department of Computer Science & Engineering
Microprocessor & Computer Architecture
UE24CS251B

MPCA Lab Programs – 7
Plug-in Programs

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1 **LED Light Up**

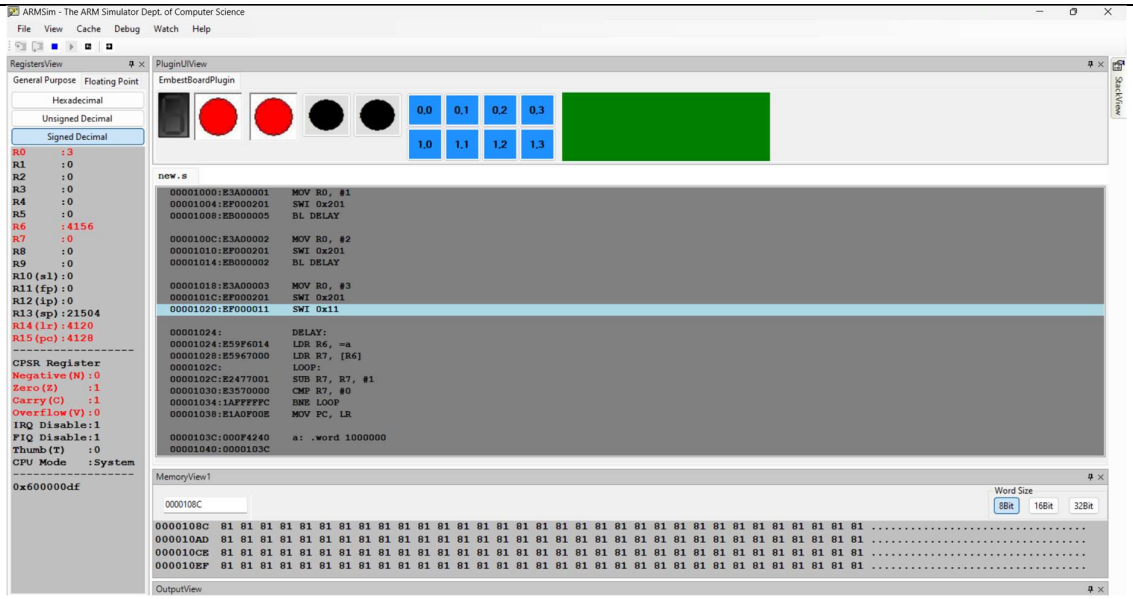
```
MOV R0, #1
SWI 0x201
BL DELAY

MOV R0, #2
SWI 0x201
BL DELAY

MOV R0, #3
SWI 0x201
SWI 0x11

DELAY:
LDR R6, =a
LDR R7, [R6]
LOOP:
SUB R7, R7, #1
CMP R7, #0
BNE LOOP
MOV PC, LR

a: .word 1000000
```



2 **Seven Segment Display**

UPCOUNTER

a: .word 1000000

.EQU A, 0X80
.EQU B, 0X40
.EQU C, 0X20
.EQU P, 0X10
.EQU D, 0X08
.EQU E, 0X04
.EQU F, 0X02
.EQU G, 0X01

TAB:
.WORD A|B|C|D|E|G ;0
.WORD B|C ;1
.WORD A|B|F|E|D ;2
.WORD A|B|F|C|D ;3
.WORD G|F|C|B ;4
.WORD A|G|F|C|D ;5
.WORD A|G|E|D|C|F ;6
.WORD A|B|C ;7
.WORD A|B|C|D|E|F|G ;8
.WORD A|B|G|F|C ;9
.WORD A|G|E|C|B|F ;A
.WORD A|B|C|D|E|F|G|P ;B
.WORD A|G|E|D ;C
.WORD A|C|D|E|G|P|B ;D

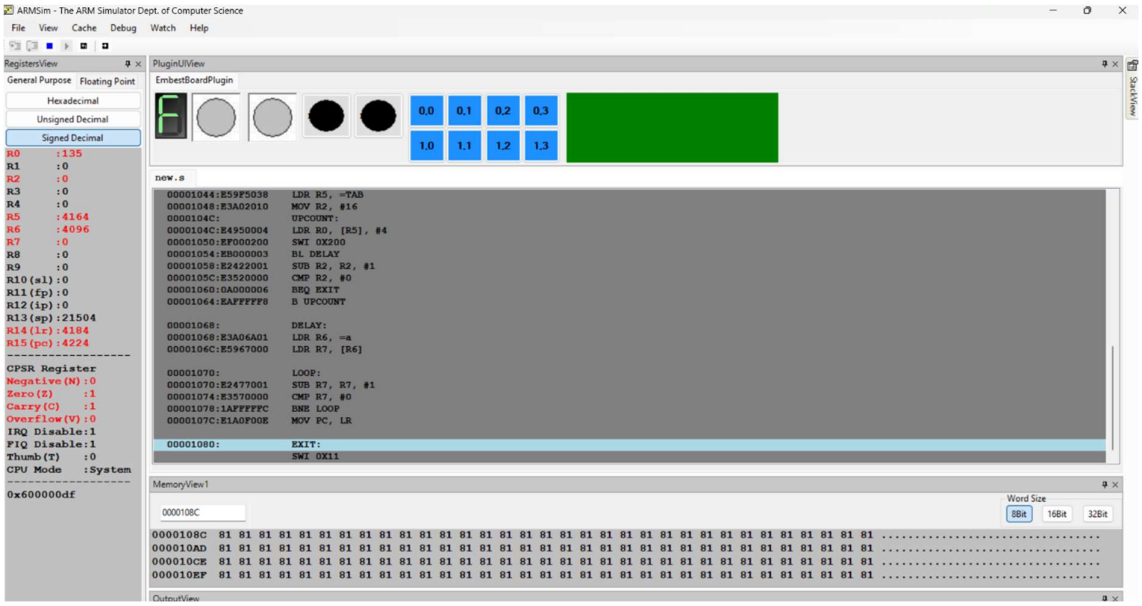
```
.WORD A|G|F|E|D      ;E
.WORD A|G|F|E        ;F

LDR R5, =TAB
MOV R2, #16
UPCOUNT:
LDR R0, [R5], #4
SWI 0X200
BL DELAY
SUB R2, R2, #1
CMP R2, #0
BEQ EXIT
B UPCOUNT

DELAY:
LDR R6, =a
LDR R7, [R6]

LOOP:
SUB R7, R7, #1
CMP R7, #0
BNE LOOP
MOV PC, LR

EXIT:
SWI 0X11
```



DOWNCOUNTER

a: .word 1000000

```
.EQU A, 0X80
.EQU B, 0X40
.EQU C, 0X20
.EQU P, 0X10
.EQU D, 0X08
.EQU E, 0X04
.EQU F, 0X02
.EQU G, 0X01

TAB:
.WORD A|B|C|D|E|G     ;0
.WORD B|C             ;1
.WORD A|B|F|E|D       ;2
.WORD A|B|F|C|D       ;3
.WORD G|F|C|B         ;4
.WORD A|G|F|C|D       ;5
.WORD A|G|E|D|C|F     ;6
.WORD A|B|C           ;7
.WORD A|B|C|D|E|F|G   ;8
.WORD A|B|G|F|C       ;9
.WORD A|G|E|C|B|F     ;A
.WORD A|B|C|D|E|F|G|P ;B
.WORD A|G|E|D         ;C
.WORD A|C|D|E|G|P|B   ;D
.WORD A|G|F|E|D       ;E
.WORD A|G|F|E         ;F

LDR R5, =TAB
ADD R5, R5, #60
MOV R2, #16

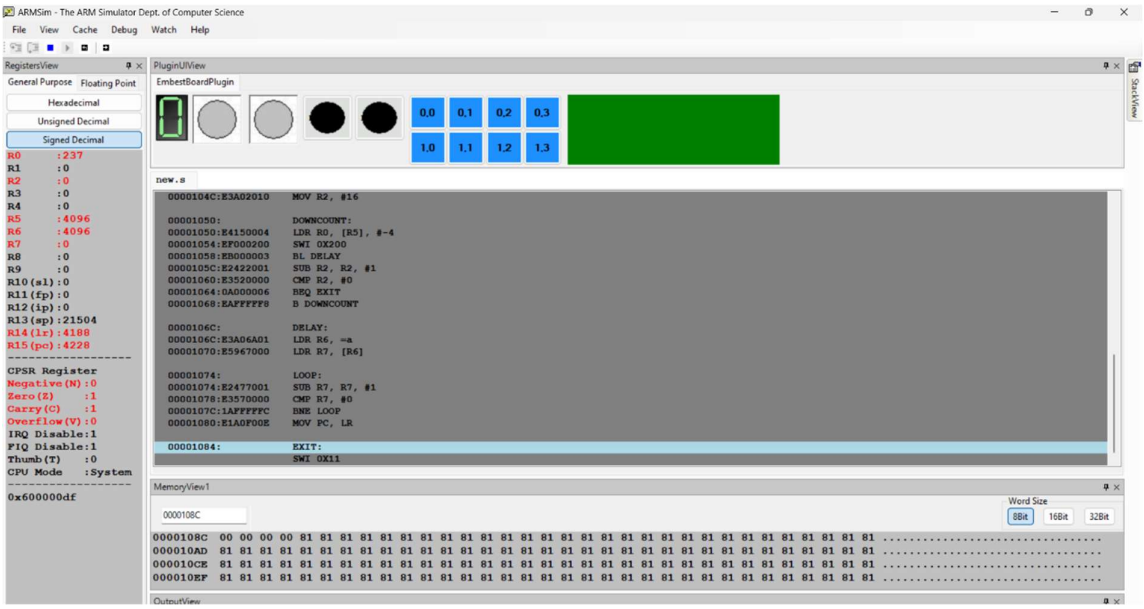
DOWNCOUNT:
LDR R0, [R5], #-4
SWI 0X200
BL DELAY
SUB R2, R2, #1
CMP R2, #0
BEQ EXIT
B DOWNCOUNT

DELAY:
LDR R6, =a
```

```
LDR R7, [R6]

LOOP:
SUB R7, R7, #1
CMP R7, #0
BNE LOOP
MOV PC, LR

EXIT:
SWI 0X11
```



UP AND DOWN COUNTER

```
a: .word 1000000

.EQU A, 0X80
.EQU B, 0X40
.EQU C, 0X20
.EQU P, 0X10
.EQU D, 0X08
.EQU E, 0X04
.EQU F, 0X02
.EQU G, 0X01

TAB:
.WORD A|B|C|D|E|G
.WORD B|C
.WORD A|B|F|E|D
```

```
.WORD A|B|F|C|D
.WORD G|F|C|B
.WORD A|G|F|C|D
.WORD A|G|E|D|C|F
.WORD A|B|C
.WORD A|B|C|D|E|F|G
.WORD A|B|G|F|C
.WORD A|G|E|C|B|F
.WORD A|B|C|D|E|F|G|P
.WORD A|G|E|D
.WORD A|C|D|E|G|P|B
.WORD A|G|F|E|D
.WORD A|G|F|E



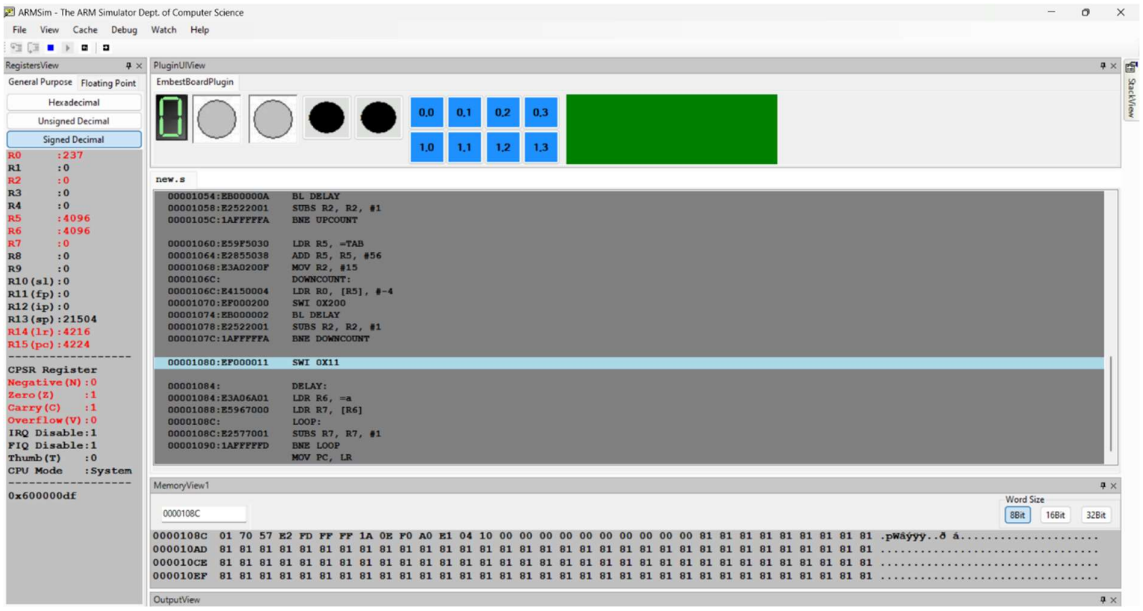
LDR R5, =TAB
MOV R2, #16
UPCOUNT:
LDR R0, [R5], #4
SWI 0X200
BL DELAY
SUBS R2, R2, #1
BNE UPCOUNT

LDR R5, =TAB
ADD R5, R5, #56
MOV R2, #15
DOWNCOUNT:
LDR R0, [R5], #-4
SWI 0X200
BL DELAY
SUBS R2, R2, #1
BNE DOWNCOUNT

SWI 0X11

DELAY:
LDR R6, =a
LDR R7, [R6]
LOOP:
SUBS R7, R7, #1
BNE LOOP
```

MOV PC, LR



3 **Move a string from Right to Left on LCD (40 columns by 15 rows)**

```
.TEXT
MOV R0 , #30  ; R0 = X
MOV R1 , #7    ; R1 = Y
MOV R7 , #0
LDR R8 , =NUM
LDR R8 , [R8]
LDR R2 , =STR

LOOP:
SWI 0X204
BL SUM
CMP R0 , #0
SUBNE R0 , R0 , #1
SWIEQ 0X11
B LOOP

SUM:
CMP R7 , R8
ADDNE R7 , R7 , #1
BNE SUM
SWI 0X206
MOV R7 , #0
MOV PC , LR

.DATA
STR:   .ASCIZ  "HELLO WORLD"
```

NUM: .WORD 15000

